

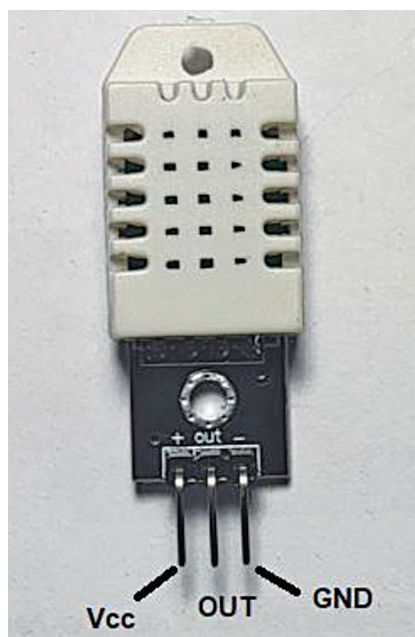


Fig. 4: Pin details of DHT22 module



Fig. 5: Relay module

Circuit and working

The circuit diagram of weather station with device control is shown in Fig. 2. It comprises Arduino Uno Board1, DHT22 temperature and humidity sensor, BC548 transistor T1, a 5V single changeover relay, light-emitting diode LED1, and resistors R1 and R2. The 6cm TFT touch-screen display to be mounted on top of the Arduino board is shown in Fig. 3.

A 9V-12V DC power supply is used to power the Arduino board. The touch-screen, DHT22 sensor, and BC548 device driver sections are powered by the 5V supply from Arduino board.

Arduino Uno. It is the heart of the project. Temperature and humidity data, TFT touch input display, and device control inputs are processed through Arduino programming.

TFT touch-screen. The 6cm TFT touch-screen display TS1 is basically

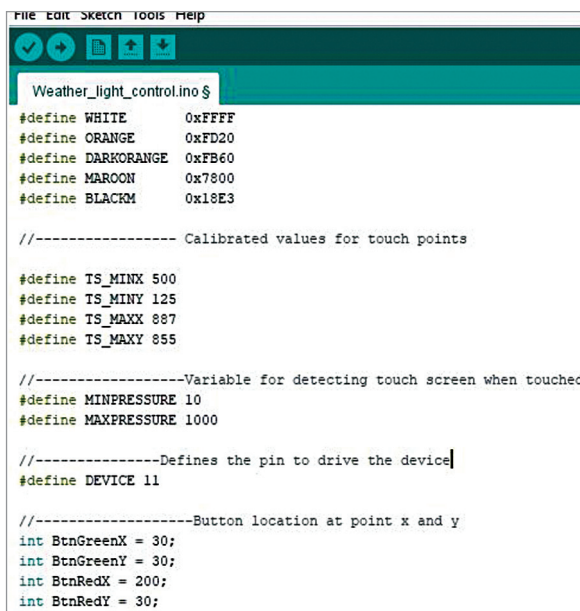


Fig. 6: The calibrated values for x and y coordinates



Fig. 7: Main console

an Arduino shield, which can be mounted on top of the Arduino Uno board. This is one of the advantages of this display because you don't need any external wires to interface with Arduino.

DHT22. This temperature and humidity sensor gives more accurate readings as compared to the popular DHT11 sensor. With this sensor, you can easily program to get humidity in %, temperature in °C and °F units. In this project, its output pin is connected to pin 11 of Arduino board.

DHT22 sensor can either be purchased as a sensor or as a module. The sensor comes as a 4-pin package out of which only three pins are used. The module comes as a 3-pin package, as shown in Fig. 4. The module has a filtering capacitor and pull-up resistor inbuilt, but for the

sensor you must use them externally, if required. Otherwise, performance of both is same. The sensor is also factory calibrated and hence easy to interface with microcontrollers. It can measure temperature from -40°C to +80°C and humidity from 0% to 100%.

BC548. This popular bipolar NPN transistor is used as driver for switching the electrical appliance connected across the relay module.

Relay module. This is an electromechanical sugar-cube, single-channel relay used to switch on/off the electrical device connected across its COM and NO terminals. You can use readily available single-channel relay module. In that case you can connect digital pin 11 of Arduino to IN pin of relay module. The pin details of relay module are shown in Fig. 5.

The working of the circuit is simple. After uploading the code to Arduino Uno board and connecting all the components, as shown in the circuit, switch on the circuit. You will see the main panel on the TFT touch-screen, as shown in Fig. 7. Touch the Device Control option, using stylus or your fingertip, to turn on or turn off the appliance/device such as tube-light or air-conditioner. Otherwise, touch on the Weather Station option to check the current temperature and humidity.

Software

The software (weather_light_control.ino) for this project is written in Arduino programming language. The project requires specific libraries for DHT22 and touch-screen display, such as DHT.h, SPFD5408_Adafruit_GFX.h, SPFD5408_Adafruit_TFTLCD.h and SPFD5408_